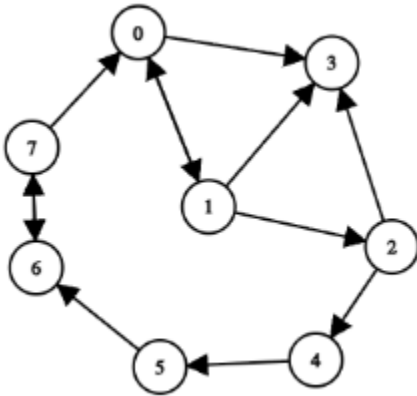


ComSci 342 Homework 5

1. [FuncLang for graphs]

a. Write a FuncLang program to represent the graph shown below:



```
(list
  (cons 7 0)
  (cons 0 3)
  (cons 0 1)
  (cons 1 0)
  (cons 1 3)
  (cons 1 2)
  (cons 2 3)
  (cons 2 4)
  (cons 4 5)
  (cons 5 6)
  (cons 6 7)
  (cons 7 6))
```

b. Write a FuncLang lambda function and assign it to a variable countedges, where given one node and a directed graph, return the number of edges that connect to or from that node.

```
(define countedges
  (lambda (node graph)
    (if (null? graph)
        0
        (+ (if (= node (car (car graph))) 1 0)
           (if (= node (cdr (car graph))) 1 0)
           (countedges node (cdr graph))))))
```

2. [FuncLang for bit-strings] Write a FuncLang lambda function and assign it to a variable hdistance, to compute Hamming Distance of two strings.

```
(define hdistanceHelper
  (lambda (a b)
    (if (null? a)
        (if (null? b)
            0
            -1)
        (if (null? b)
            -1
            (let ((rest (hdistanceHelper (cdr a) (cdr b))))
              (if (= rest -1)
                  -1
                  (+ (if (= (car a) (car b)) 0 1)
                     rest)))))))

(define hdistance
  (lambda (a b)
    (hdistanceHelper a b)))
```

3. [FuncLang for numbers]

- a. Write a FuncLang lambda function and assign it to a global variable func, that computes $f(x) = 2x^2 - x + 1$ given x.

```
(define func
  (lambda (x)
    (+ (- (* 2 (* x x)) x) 1)))
```

- b. Write the FuncLang lambda function which accepts 3 parameters as input (x, f, z) and assign it to a variable Generate, whose functionality is to produce the list of pairs (x f(x)) as described below.

\$ (Generate 1 func 3)

((1 2) (2 7) (3 16))

\$ (Generate 1 func 0)

()

\$ (Generate 1 func -1)

()

\$ (Generate 2 func 1)

((2, 7))

```
(define Generate
  (lambda (x f z)
    (if (= z 0)
        (list)
        (if (< z 0)
            (list)
            (cons (list x (f x))
                  (Generate (+ x 1) f (- z 1)))))))
```